Lesson 2.2: Refining your Model with GenAI

Location: Classroom

Grade Level: High School

Time: 60 minutes

Driving Question
- How will climate change affect what we value about Orange County's ocean?

Lesson Synopsis
In this 60-minute lesson, student teams refine their initial models of how climate change impacts the values Orange County communities place on the ocean.

Students begin by analyzing their initial model to identify what questions they have and to identify information gaps in the model. They then use generative AI (GenAI) chatbots to gather additional information on other perspectives, ideas, and data that helps to address the gaps in their models. Finally, student teams use the gathered information and integrate the new perspectives into their initial model.

What students will figure out:
- Climate change affects ocean systems, which in turn influences the values people place on them.
- As ocean conditions change, communities experience shifts in the values they place on the ocean, such as its emotional, economic, or cultural significance.
- To best understand how these changes will impact the community, it is helpful to communicate with experts in the field, though this is not always simple.
- Using AI tools can help us explore new perspectives, identify gaps in our understanding, and refine our models to better represent these complex relationships.

LEARNING OUTCOMES

By the end of this investigation, students will be able to…	You can assess this using…
1. Identify areas within their model that they want to elaborate on or expand upon.	Student discussions in the *Problematize* section
2. Develop a plan and refine their SageModeler models further by incorporating sourced information from the AI chatbots.	Student investigation in the *Investigate* section
3. Reflect on how they can further investigate and use real-world data to improve their models	Student discussion in the *Problematize* section

Building towards NGSS and AI Literacy:

- **HS-ESS3-6:** Use a computational representation to illustrate the relationships among Earth systems and how those relationships are being modified due to human activity. Students are refining models based on interaction with the environment and the cyclical effects that occur as a result.
- **AI Literacy's Big Idea #4**: **Natural Interaction**. Describe several approaches to natural language processing (e.g., how a chatbot can work using keywords, templates, or machine learning like GenAI). In this lesson, students will interact with several GenAI chatbots. They will develop prompts for the GenAI chatbots to help incorporate additional stakeholders' perspectives and improve their models.
- **AI Literacy's Big Idea #5: Ethical AI.** Articulate how to design an AI system in ethical ways (e.g., incorporating multiple stakeholders' perspectives and values). In this lesson, students will reflect on the interactions with the GenAI chatbots (e.g., whether the responses are helpful, accurate, and relevant) and articulate the advantages or challenges of using these tools.

LESSON OVERVIEW

Section	Time	What Happens
Problematize	10 minutes	Students are invited to revisit their initial SageModeler models, which depict how varying components, processes, and interactions impact their communities' value to the ocean. They reflect on the initial relationships they set and consider areas where they might strengthen their models by gathering additional information.
Investigate	35 minutes	Students work in small teams to brainstorm areas within their models that may need further exploration or refinement. They consider potential gaps or missing data that could improve the accuracy of their models. Each team identifies specific aspects of their model to investigate further and uses GenAI to gather relevant information. They focus on how climate change may affect these aspects of the model, such as changes in ocean temperature, rising sea levels, or shifts in coastal ecosystems.

		After gathering information, student teams revisit their initial models and incorporate their newfound insights to refine how climate change impacts the value placed on the ocean.
Putting the pieces together	10 minutes	Students share their experiences of using AI to help support their model revisions and highlight new additions to their model. Students start to reflect on other ways to gather more information, data sources, to further enhance their models.
Problematize	5 minutes	Students begin reflecting on how they can use real-world data to validate relationships within their models and identify additional areas to research.

MATERIALS

- Lesson 2.2 slideshow
- Science Journal
- Module 2.1: SageModeler Instructions (Student Facing; in science journal as well)
- A screen to view the slideshow with audio (projector, Promethean, etc.)

BEFORE YOU START TEACHING

1. Get acquainted with AI Chatbot Platform. Make sure the site is accessible on the students' devices.

2. Review Student SageModeler Instructions to model or support students.

3. Include a numbered list of general steps to set up the investigation before students arrive.

4. This can also include other things that educators should do or think about before starting the lesson, such as checking for safety concerns.

WHERE WE ARE GOING AND NOT GOING

Where We Are Going: This lesson focuses on using AI to help refine the models students have built in the previous lesson. They will use AI chatbots to help identify or strengthen the causal relationships between biotic, abiotic, human value, and human impact components in their models. Beyond simple linear links, students are encouraged to think about **changes over time**, **feedback loops (i.e., nonlinear interactions),** and **causal chains** linking components. The chatbots will also help students identify uncertainty in their models (where students might want to conduct additional research) to prepare them for model revisions.

- **HS-ESS3-6:** Students are refining models based on interaction with the environment and the cyclical effects that occur as a result.
- **AI Literacy's Big Idea #4**: Natural Interaction. Students will develop prompts for the generative AI chatbots to improve their models following the CLEAR framework that they encountered in Module 1. Students will gain an understanding of how generative AI chatbots respond to prompts in natural language.
- **AI Literacy's Big Idea #5**: Ethical AI. Students will reflect on the advantages and challenges of using generative AI in constructing models and refining science investigations.

Where We Are NOT Going: Students will be focusing on local community values and their shifts with climate change. They will not be incorporating global impacts and global values into their models. This lesson will also emphasize how using AI can help, but it has drawbacks. Students will **not** use AI chatbots as the only information-gathering source. Rather, these tools are the starting point to get students to think more critically about areas to improve in their models.

LEARNING SEQUENCE
Lesson plan notes are also provided on the slide notes.

PROBLEMATIZE: **Introducing Today's Session (10 minutes)**

1. Open the slideshow to **Slide 1** and welcome students into the class and tell them that today, they will continue working on their teams to refine the models that they constructed and got feedback on in the last session.

2. Before beginning, advance the slideshow to **Slide 2** and ask student teams to reflect on their team's model so far by answering the following questions.
   - Are there areas of your model that you have questions or are unsure about?
   - Are there sections of your model that need more information?
   - How might your team gather this information?

3. After student teams have reflected and written down their questions for their model, tell students that when scientists create models, they constantly look at ways to improve their models to make sense of the system. However, this means finding ways of gathering new information.

4. Advance to **Slide 3** to give students an overview of what they will do in lesson 2.2.

5. Afterwards, move on to **Slide 4** and bring the whole class together as a group and discuss the following questions:
   - How have you collected information for your models so far? How did this information contribute to your model?
   - How else may a scientist collect information?

6. After students share their responses, reaffirm ideas that they have gathered information by searching for themselves and interviewing community members. Today, they will investigate another method of information gathering: using AI.

INVESTIGATE: **Prepping the Chatbot Interview (15 minutes)**

1. Advance to **Slide 5,** where students will see an image of the AI chatbot platform with the 5 AI chatbots that are available to chat with and a default ChatGPT.

2. Share with students that the chatbot characters they see on the screen were created by feeding them information that was gathered by conducting community interviews, just like the ones that they conducted.

3. Advance to **Slide 6** and tell students that they will "interview" these chatbots to help them gather more information to strengthen their models, but just like a real interview, they will need to prepare.

4. Start by telling students that you would like them to just explore the chatbots.

5. Once the time is up, bring the class together to explore the following questions
   - What did you try asking or doing with the chatbots?
   - What types of questions did it seem to handle well?
   - What surprised you about how the chatbot responded?

6. Next, move on to **Slide 7**. To help the students prepare, ask students to open their Science Journal to **Module 2.2: Refining your Model with GenAI (Student Task: Preparing your Chatbot Interviews)** **p.20-21**, and tell them that just like before, they will have to prepare their interview questions in advance, but unlike last time, they will have five experts to chat with.

What questions do you have about your model?	**Who can help you gather more information to answer your question?**	**After talking to the chatbots, what do you want to add to your model?**
What are some measurable components that contribute to ocean acidification that I might include in my model?	❑ Kelp Biologist ❑ Fisherman ❑ Influencer ❑ Student ❑ Civil Engineer	

7. Advance to **Slide 8** and tell students that each member in their team will be responsible for investigating a question with 1-2 chatbots and bringing back the information to adjust their model. If students need support in writing questions, ask the following:
   - What parts of your model feel uncertain or incomplete?
   - Are there any relationships you are unsure about?
   - Are there other relationships between components that you want to learn more about?

8. Once students have their questions ready, tell students to select which chatbot may know more about their question.

   Remind them that their interview with AI may be a little different from one with a person.

9. Advance to **Slide 9** to share with students some tips to keep in mind when interviewing the chatbots.
   - Give context to the chatbot and frame your questions clearly.
   - Avoid asking too many questions in one prompt.
   - Ask follow-up questions
   - Recognize that AI has boundaries and that it is not human

INVESTIGATE: Conducting the Chatbot Interview (20 minutes)

1. Make sure that each student has a question that they want to investigate and a computer that they can participate with, as they will be working individually to start.

2. When students are ready to begin, pull out the instructions on **Slide 10** to conduct the chatbot interview.
   - Visit the AISciComm chatbot platform and choose the chatbot that you want to interview for your question.
   - Upload the snapshot of your team's model that you took back in module 2.1. If you haven't done so yet, go back to your SageModeler model and take a snapshot.
   - When the image loads, give the chatbot context for what you are uploading. You can give the following prompt for students to fill out.
     - ***Hello, my team and I are [grade level] students in [class]. We are working on a model that demonstrates how changes to the climate impact ocean systems and affect [your chosen value]. Can you describe the model to me?***
   - Share the question you are investigating with the chatbot. Make sure to use follow-up questions.
   - If you have identified other bots that may know more about your question, make sure to interview them too.
   - Remember to take notes!

3. Once students are ready, give them time to individually chat with the chatbots to help them answer their questions.

4. As students converse with the chatbots, make sure that they are taking notes on the information that they want to share back with the group. Remind students to make sure they are filling out the interview document in their Science Journal.

5. After students have had time to chat with the chabots, advance to **Slide 11** and ask student teams to get back together to share their findings with their team and fill out the last section on their chatbot interview document together, if they haven't done so already

6. Advance to **Slide 12** and give students a moment to revisit their model from the previous session and make modifications to the model, based on their conversation with the chatbots.

PUTTING THE PIECES TOGETHER: **Elaborating on our GenAI Experience (10 minutes)**

1. Advance to **Slide 13**. Once the student teams finish up the revisions to their model, bring the class back together for a class conversation. Ask students the following questions
    - What were some of the most surprising findings you gained from your chatbot interviews?
    - Did the chatbots provide new ideas or connections that you hadn't thought of before?
    - What revisions did you incorporate into your model?

2. Advance to **Slide 14**. As students share their experiences, ask the following questions:
    - Looking at your current model, what specific data is still missing that could help you better understand the system you're modeling?
    - Where might we gather or collect this data?

PROBLEMATIZE: **Reflecting on Module 2 (5 minutes)**

1. As you wrap up for the day, share with students that their final task for the day is to reflect on the process of refining their model.

2. Advance to **Slide 15** and have student teams work together to answer the following questions together on **Module 2.2: Refining your Model with GenAI (Student Reflection), p. 22** of their Science Journals.
   - How did the AI chatbots help change your thinking and support the changes in your model?
   - Which areas of your model need more information? How might you explore them further?

3. As students wrap up their discussion, and if time allows, have teams share with the entire class.

4. Thank the students for their time today. Let students know that during the next session, they are going to take more time to refine their models by exploring how to incorporate real-world data to validate the relationships within their models.

EXTEND (OPTIONAL TAKE-HOME ACTIVITY)

Time: 15 minutes **Goal:** ● Critically evaluate responses generated by GenAI ● Reflect on how to prompt GenAI **Notes:**	**Extend: Evaluate AI's responses (15 minutes; take-home activity)** 1. Students review their notes, and then reflect on (1) the prompts they sent to the chatbots and (2) the chatbots' generated responses. Students answer the questions on **Module 2.2: Refining your Model with GenAI (Optional: Reflecting on AI's Responses to your Questions), p. 23-24** of their Science Journals: ● Did your team ask questions that followed the CLEAR framework (Concise, Logical, Explicit, Adaptive, Reflective)? How did your questions reflect this? If not, why? ● Did the chatbots provide answers that helped you learn more about your question? What kind of information or examples did they give? How did this connect to your community, social networks, local knowledge, or experiences? ● Were the chatbot responses relevant to your question? Did you notice any inaccuracies or missing information? Give specific examples.

- GenAI chatbots learn by adjusting patterns from the data they are trained on, while traditional search engines retrieve existing information. How does understanding this difference change the way you use the chatbots? How did this influence your model updates?
- What ideas would you give to improve the chatbots' answers? How could the chatbots better support your learning about climate change and the ocean, while reflecting your community context and assets?

